
Problem 1

evaluate line integral using analytical, simpson's, trapezoidal, and quad functions

$$\int_0^1 x^{0.1}(1.2-x)(1-e^{20(x-1)}) dx$$

```
function ass2_1
% analytical evaluation of line integral
syms x;
I_analytical= double(int((x^0.1)*(1.2-x)*(1-exp(20*(x-1))),0,1));
clear x;
% I_analytical=0.602297;
f=@exp_func; a=0; b=1; n=1000;
% simpsons evaluation of integral
I_simpson = simpson(f,a,b,n);
% trapezoidal evaluation
I_trapezoidal = trapezoidal(f,a,b,n);
% MATLAB quad function (uses recursive simpson's 1/3 rule)
I_matlab=integral(f,a,b);
fprintf('The analytical value of line integral is %g\n',I_analytical);
fprintf('The trapezoidal value of line integral = %g\n',I_trapezoidal);
error=100*abs(I_trapezoidal-I_analytical)/abs(I_analytical);
fprintf('Percent error of trapezoidal solution = %g %%\n\n',error);
fprintf('The simpson value of line integral = %g\n',I_simpson);
error=100*abs(I_simpson-I_analytical)/abs(I_analytical);
fprintf('Percent error of simpson solution = %g %%\n\n',error);
fprintf('The quad function value of line integral = %g\n',I_matlab);
error=100*abs(I_matlab-I_analytical)/abs(I_analytical);
fprintf('Percent error of integral function solution = %g %%\n',error);
return
%
% trapezoidal line integral function
function [I_trap]= trapezoidal(f,a,b,n)
I_trap=0;
h=(b-a)/n;
for i=0:n
    x=a+i*h;
    if (i == 0 || i == n) % end points
        I_trap=I_trap+feval(f,x);
    else % interior points
        I_trap=I_trap+2*feval(f,x);
    end
end
I_trap=I_trap*h/2;
return
%
% simpson line integral function for even number of segments (1/3 rule)
```

```

function [I_simp] = simpson(f,a,b,n)
I_simp = 0;
x=0;
h=(b-a)/n;
for i=0:n
    x=a+i*h;
    if (i == 0 || i == n) % end points
        I_simp=I_simp+feval(f,x);
    elseif (mod(i,2) == 0) % even
        I_simp=I_simp+2*feval(f,x);
    else % odd
        I_simp=I_simp+4*feval(f,x);
    end
end
I_simp=I_simp*h/3;
return
%
function [f] = exp_func(x)
f=(x.^0.1).*(1.2-x).*(1-exp(20*(x-1)));
return
%
```

The analytical value of line integral is 0.602298

*The trapezoidal value of line integral = 0.602047
Percent error of trapezoidal solution = 0.041712 %*

*The simpson value of line integral = 0.602143
Percent error of simpson solution = 0.0257819 %*

*The quad function value of line integral = 0.602298
Percent error of integral function solution = 6.21152e-07 %*

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Problem 2

Moment on a beam

$$M = \int_0^{11} 5 + 0.25x^2 dx$$

```
function ass2_2
f=@moment_func;
n=24; a=0; b=12;
syms x;
I_anal= double(int(5+0.25*x^2,0,12));
I_trap = trapezoidal(f,a,b,n);
I_simp = simpson(f,a,b,n);
fprintf('The trapezoidal value of moment = %g\n',I_trap);
fprintf('The simpson value of moment = %g\n',I_simp);
fprintf('The analytical value of moment = %g\n',I_anal);
return

function [M] = moment_func(x)
M=5+0.25*x^2;
return

%
% trapezoidal line integral function
function [I_trap]= trapezoidal(f,a,b,n)
I_trap=0;
h=(b-a)/n;
for i=0:n
    x=a+i*h;
    if (i == 0 || i == n) % end points
        I_trap=I_trap+feval(f,x);
    else % interior points
        I_trap=I_trap+2*feval(f,x);
    end
end
I_trap=I_trap*h/2;
return

%
% simpson function for both odd and even number of segments
% 1/3 rule for first n-3 segments and 3/8 rule for last 3 segments
function [I_simp] = simpson(f,a,b,n)
I_simp = 0;
h=(b-a)/n;
if (mod(n,2) == 1) % n is odd
    n=n+1;
end
for i=0:n
    x=a+i*h;
    if (i == 0 || i == n) % end points
        I_simp=I_simp+feval(f,x);
    elseif (mod(i,2) == 0) % even
        I_simp=I_simp+2*feval(f,x);
```

```
        else % odd
            I_simp=I_simp+4*feval(f,x);
        end
    end
    I_simp=I_simp*h/3;
    return
%
```

The trapezoidal value of moment = 204.125
The simpson value of moment = 204
The analytical value of moment = 204

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%

Problem 3

sailboat problem using simpson's, trapezoidal, and quad functions

$$F = \int_0^H \frac{200z}{7+z} e^{2.5z/H} dz$$

$$d = \frac{\int_0^H \frac{200z^2}{7+z} e^{2.5z/H} dz}{F}$$

```
function ass2_3
f1=@force; f2=@force_moment;
a=0; b=30; n=10;
% simpsons evaluation of integral
F_simpson = simpson(f1,a,b,n);
d_simpson = simpson(f2,a,b,n)/F_simpson;
fprintf('The simpson value of force = %g\n',F_simpson);
fprintf('The simpson value of distance = %g\n',d_simpson);
% trapezoidal evaluation
F_trapezoidal = trapezoidal(f1,a,b,n);
d_trapezoidal = trapezoidal(f2,a,b,n)/F_trapezoidal;
fprintf('The trapezoidal value of force = %g\n',F_trapezoidal);
fprintf('The trapezoidal value of distance = %g\n',d_trapezoidal);
% MATLAB integral function
F_matlab=integral(f1,a,b);
d_matlab=integral(f2,a,b)/F_matlab;
fprintf('The integral function value of force = %g\n',F_matlab);
fprintf('The integral function value of distance = %g\n',d_matlab);
return
%
function [I_trap]= trapezoidal(f,a,b,n)
I_trap=0;
h=(b-a)/n;
for i=0:n
    x=a+i*h;
    if (i == 0 || i == n) % end points
        I_trap=I_trap+feval(f,x);
    else % interior points
        I_trap=I_trap+2*feval(f,x);
    end
end
I_trap=I_trap*h/2;
return

function [I_simp] = simpson(f,a,b,n)
I_simp = 0;
h=(b-a)/n;
for i=0:n
    x=a+i*h;
```

```

        if (i == 0 || i == n) % end points
            I_simp=I_simp+feval(f,x);
        elseif (mod(i,2) == 0) % even
            I_simp=I_simp+2*feval(f,x);
        else % odd
            I_simp=I_simp+4*feval(f,x);
        end
    end
    I_simp=I_simp*h/3;
    return

function [f] = force(z)
H=30;
f=200*(z./(7+z)).*exp(-2.7*z/H);
return

function [f] = force_moment(z)
H=30;
f=200*z.*(z./(7+z)).*exp(-2.7*z/H);
return

The simpson value of force = 968.357
The simpson value of distance = 12.1556
The trapezoidal value of force = 949.005
The trapezoidal value of distance = 12.3729
The integral function value of force = 970.453
The integral function value of distance = 12.1164

```

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Problem 4

Flowrate Q through a pipe of Radius R

$$Q = 4\pi \int_0^R r(1 - r/R)^{1/6} dr$$

```
function ass2_4
% analytical evaluation of line integral
syms r;
Q_analytical= double(int(4*pi*r*(1-r/2)^(1/6),0,2));
fprintf('The analytical value of Q is %g\n',Q_analytical);
f=@pipe_func; a=0; b=2; n=10;
% simpsons evaluation of integral
Q_simpson = simpson(f,a,b,n);
fprintf('The simpson value of Q = %g\n',Q_simpson);
error=100*abs(Q_simpson-Q_analytical)/abs(Q_analytical);
fprintf('Percent error of simpson solution = %g %%\n',error);
%
% trapezoidal evaluation
Q_trapezoidal = trapezoidal(f,a,b,n);
fprintf('The trapezoidal value of Q = %g\n',Q_trapezoidal);
error=100*abs(Q_trapezoidal-Q_analytical)/abs(Q_analytical);
fprintf('Percent error of trapezoidal solution = %g %%\n',error);
%
% MATLAB integral function (uses recursive simpson's 1/3 rule)
Q_integral=integral(f,a,b);
fprintf('The integral function value of Q = %g\n',Q_integral);
error=100*abs(Q_integral-Q_analytical)/abs(Q_analytical);
fprintf('Percent error of integral function solution = %g %%\n',error);
return
%

function [I_trap]= trapezoidal(f,a,b,n)
I_trap=0;
h=(b-a)/n;
for i=0:n
    x=a+i*h;
    if (i == 0 || i == n) % end points
        I_trap=I_trap+feval(f,x);
    else % interior points
        I_trap=I_trap+2*feval(f,x);
    end
end
I_trap=I_trap*h/2;
return

function [I_simp] = simpson(f,a,b,n)
I_simp = 0;
h=(b-a)/n;
for i=0:n
```

```

    x=a+i*h;
    if (i == 0 || i == n) % end points
        I_simp=I_simp+feval(f,x);
    elseif (mod(i,2) == 0) % even
        I_simp=I_simp+2*feval(f,x);
    else % odd
        I_simp=I_simp+4*feval(f,x);
    end
end
I_simp=I_simp*h/3;
return

```

```

function [f] = pipe_func(r)
R=2;
f=4*pi*r.*(1-r/R).^(1/6);
return

```

The analytical value of Q is 19.8852
The simpson value of Q = 19.1392
Percent error of simpson solution = 3.75172 %
The trapezoidal value of Q = 18.5939
Percent error of trapezoidal solution = 6.49379 %
The integral function value of Q = 19.8852
Percent error of integral function solution = 3.8261e-07 %

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Problem 5

flow in a river

$$A_c = \int_0^b H(y) dy \quad \bar{H} = \frac{A_c}{b} \quad Q = \int_0^b U(y)H(y) dy \quad \bar{U} = \frac{Q}{A_c}$$

```
function ass2_5
y = [0, 1, 3, 5, 7, 8, 9, 10];
H = [0, 1, 1.5, 3, 3.5, 3.2, 2, 0];
U = [0, 0.1, 0.12, 0.2, 0.25, 0.3, 0.15, 0];
n=length(y);
I1=0; I2=0;
for i=1:n-1
    dy=y(i+1)-y(i);
    I1=I1+dy*(H(i)+H(i+1))/2;
    I2=I2+dy*(H(i)*U(i)+H(i+1)*U(i+1))/2;
end
b = max(y);
Ac = I1;
H_bar = I1/b;
Q = I2;
U_bar = Q/Ac;
fprintf('Cross sectional area = %g m^2\n',Ac);
fprintf('Average depth = %g m\n', H_bar);
fprintf('Flow rate = %g m^3/s\n',Q);
fprintf('Average velocity = %g m/s\n',U_bar);
return
%
```

```
Cross sectional area = 20.95 m^2
Average depth = 2.095 m
Flow rate = 4.2825 m^3/s
Average velocity = 0.204415 m/s
```

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Problem 6

Double integration example

$$\int_{-2}^2 \int_0^4 (x^2 - 3y^2 + xy^3) dx dy$$

```
function ass2_6
syms x y;
I_analytical=double(int(int(x^2-3*y^2+x*y^3,x,-2,2),y,0,4));
clear x y;
n=10;
ax=-2;
bx=2;
ay=0;
by=4;
f=@xyfunc;
I_trap = double_trapezoidal(f,ax,bx,ay,by,n);
fprintf('analytical value of double integral %g\n',I_analytical);
fprintf('trapezoidal value of double integral %g\n',I_trap);
I_simp = double_simpson(f,ax,bx,ay,by,n);
fprintf('simpson value of double integral %g\n',I_simp);
I_matlab = integral2(f,ax,bx,ay,by);
fprintf('matlab integral function value of double integral %g\n',I_matlab);
return
%

function [I_trap]= double_trapezoidal(f,ax,bx,ay,by,n)
I_trap=0;
hy=(by-ay)/n;
for i=0:n
    y=ay+i*hy;
    if (i == 0 || i == n) % end points
        I_trap=I_trap+feval('trapezoidal',f,ax,bx,y,n);
    else % interior points
        I_trap=I_trap+2*feval('trapezoidal',f,ax,bx,y,n);
    end
end
I_trap=I_trap*hy/2;
return

function [I_trap]= trapezoidal(f,a,b,y,n)
I_trap=0;
h=(b-a)/n;
for i=0:n
    x=a+i*h;
    if (i == 0 || i == n) % end points
        I_trap=I_trap+feval(f,x,y);
    else % interior points
        I_trap=I_trap+2*feval(f,x,y);
    end
end
I_trap=I_trap*h/2;
```

```

        end
    end
    I_trap=I_trap*h/2;
    return

function [I_simp] = double_simpson(f,ax,bx,ay,by,n)
I_simp = 0;
hy=(by-ay)/n;
for i=0:n
    y=ay+i*hy;
    if (i == 0 || i == n) % end points
        I_simp=I_simp+feval('simpson',f,ax,bx,y,n);
    elseif (mod(i,2) == 0) % even
        I_simp=I_simp+2*feval('simpson',f,ax,bx,y,n);
    else % odd
        I_simp=I_simp+4*feval('simpson',f,ax,bx,y,n);
    end
end
I_simp=I_simp*hy/3;
return

function [I_simp] = simpson(f,a,b,y,n)
I_simp = 0;
x=0;
h=(b-a)/n;
for i=0:n
    x=a+i*h;
    if (i == 0 || i == n) % end points
        I_simp=I_simp+feval(f,x,y);
    elseif (mod(i,2) == 0) % even
        I_simp=I_simp+2*feval(f,x,y);
    else % odd
        I_simp=I_simp+4*feval(f,x,y);
    end
end
I_simp=I_simp*h/3;
return

function [f] = xyfunc(x,y)
f=x.^2-3*y.^2+x.*y.^3;
return

analytical value of double integral -234.667
trapezoidal value of double integral -235.52
simpson value of double integral -234.667
matlab integral function value of double integral -234.667

```

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Problem 7

evaluate triple integral ass 2.7

```
function ass2_7
% function name to integrate
f=@fxyz;
% need to input the limits as characters as we want to
% be able to handle function names as limits as well
a=[-4,0,-1]; % lower limit of integral
b=[4,6,3]; % upper limit of integral
n=10; % number of simpson segments
I3_simpson=msimp(f,a,b,n);
fprintf('The simpson value of triple integral is %g\n',I3_simpson);
% solution using triple quad
f=@fxyz_integral3;
I3_matlab=integral3(f,-4,4,0,6,-1,3);
fprintf('The matlab value triple integral is %g\n',I3_matlab);
%
% analytical evaluation of triple integral
syms x y z;
I3_analytical=double(int(int(int(z*(x^3-2*y*x-
z),x,-4,4),y,0,6),z,-1,3));
fprintf('The analytical value of triple integral is %g
\n',I3_analytical);
return
```

```
function [f] = fxyz(x)
% function f=z*(x^3-2yx)
% use notation x = x(1); y=x(2); z=x(3) etc...
f=x(3)*(x(1)^3-2*x(2)*x(1)-x(3));
return
```

```
function [f] = fxyz_integral3(x,y,z)
% function f=x^3-2yx
f=z.*(x.^3-2*y.*x-z);
return
```

```
% multi-dimensional simpson main calling function
function [I] = msimp(f,a,b,n)
m=length(a); x=zeros(1,m); k=1;
I=ksimp(f,x,a,b,n,k);
return
```

```
% recursive simpson function
function [I] = ksimp(f,x,a,b,n,k)
I=0; m=length(a);
h=(b(k)-a(k))/n;
for i=0:n
    x(k)=a(k)+i*h;
    if (k == m)
```

```

        if (i == 0 || i == n)
            I=I+feval(f,x);
        elseif (mod(i,2) == 1)
            I=I+4*feval(f,x);
        elseif (mod(i,2) == 0)
            I=I+2*feval(f,x);
        end
    else
        if (i == 0 || i == n)
            I=I+feval(@ksimp,f,x,a,b,n,k+1);
        elseif (mod(i,2) == 1)
            I=I+4*feval(@ksimp,f,x,a,b,n,k+1);
        elseif (mod(i,2) == 0)
            I=I+2*feval(@ksimp,f,x,a,b,n,k+1);
        end
    end
end
I=I*h/3;
return

```

The simpson value of triple integral is -448
The matlab value triple integral is -448
The analytical value of triple integral is -448

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Problem 8

Triple integral using Monte-Carlo

$$\int_{-1}^3 \int_0^6 \int_{-4}^4 z(x^3 - 2yx) dz dy dx$$

```
function ass2_8
% evaluate triple integral using Montecarlo integration
% analytical evaluation of triple integral
syms x y z;
I3a=double(int(int(int(z*(x^3-2*y*x-z),x,-4,4),y,0,6),z,-1,3));
clear x y z;
% integration limits as vectors in x,y,z order
a=[-4 0 -1]'; % lower limit of integral
b=[4 6 3]'; % upper limit of integral
n=1000; % number of montecarlo points
m=3; % number of dimensions = 3
% function name to integrate
f=@func;
I3n=monte(f,a,b,m,n);
error=abs(I3n-I3a);
fprintf('The analytical value of triple integral is %g\n',I3a);
fprintf('The monte-carlo value of triple integral is %g\n',I3n);
fprintf('Absolute error of montecarlo solution %g\n',error);
return
%

% montecarlo integration function
function [Im] = monte(f,a,b,m,n)
x=zeros(m,n);
mu=prod(b-a);
y=rand(m,n);
for i=1:m
    x(i,:)=a(i)+(b(i)-a(i))*y(i,:);
end
Im=mu*sum(feval(f,x))/n;
return

% function f=x^3-2yx for montecarlo integration
% input x,y,z as row vectors in x(m,n)
% use notation x = x(1,:); y=x(2,:); z=x(3,:) etc...
function [fx] = func(x)
fx=x(3,:).*(x(1,:).^3-2*x(2,:).*x(1,:)-x(3,:));
return
```

*The analytical value of triple integral is -448
The monte-carlo value of triple integral is -403.803
Absolute error of montecarlo solution 44.1974*

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